

Bryson Howell

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Summary

Computer Science student and software engineer building full-stack, systems, and native Apple applications. Experience includes Java, Python, C#, C++, Swift, SwiftUI, React, Spring Boot, REST APIs, and offline-first application design. Strong focus on performance engineering, debugging, automation, reliable data workflows, and turning real repair-industry problems into practical software products.

Education

Western Governors University

Bachelor of Science in Computer Science

Expected 2027

Experience

Lead Advanced Repair Agent

Aug 2025 – Present

Best Buy, Geek Squad

- Resolve escalated hardware, operating system, application, networking, security, and performance issues on customer computers through structured diagnostics and root-cause analysis.
- Built C# Windows tools that orchestrate Python, PowerShell, and batch scripts, reducing Windows 11 idle RAM from roughly 6–7 GB to 1.5–2.5 GB.
- Increased personal repair throughput to 5–8 devices per hour versus a team target of 1–2 by standardizing repeatable diagnostics and automation.

Lead Technician / Assistant Manager

Feb 2022 – Aug 2025

uBreakiFix

- Diagnosed hardware, software, operating system, and networking failures across computers and mobile devices while communicating solutions to non-technical customers.
- Prioritized repair queues, handled escalations, supported technicians, and balanced technical quality with customer and operational requirements.

Projects

RepairLens | Swift, SwiftUI, SwiftData, Vision, CloudKit, Core ML

- Developing an offline-first iOS and macOS diagnostics platform for repair technicians to identify devices, document damage, and manage structured repair records.
- Designing on-device OCR, barcode scanning, device recognition, and image-processing workflows using Vision, Core ML, AVFoundation, and privacy-preserving local storage.
- Architecting SwiftData persistence, CloudKit synchronization, conflict handling, accessibility, and future RepairFlow API integration for unreliable-network environments.

Dynasty Draft | [GitHub](#) | Java, Spring Boot, React, TypeScript, PostgreSQL, Docker

- Built a full-stack roster builder and six-event esports season simulator with a React frontend, Spring Boot REST API, and relational data layer.
- Modeled 100+ players, historical ratings, chemistry, and title fit to generate teams, standings, leaderboards, and 85+ simulated matches per season.
- Packaged the frontend, API, Flyway migrations, and seed data into a standalone Java 21 employer demo and Docker-based development environment.

Apple Silicon Performance Lab | Swift, SwiftUI, C++, Metal, SwiftData, Instruments

- Developing a native macOS tool for repeatable CPU, memory, storage, multithreading, and CPU-versus-GPU performance experiments on Apple silicon.
- Designing a C++ benchmark engine and Metal compute workloads with warm-ups, repeated trials, correctness checks, cancellation, and raw-sample preservation.
- Calculating latency percentiles, variability, speedup, and scaling efficiency while using Instruments, OSLog, and signposts for evidence-based profiling.

TransferDesk | C#, .NET 8, WPF, Robocopy

- Built a Windows file-migration application with queued transfers, responsive progress updates, validation, detailed logs, and modular transfer engines.
- Implemented standard, parallel, Robocopy, and archive modes to support different data sizes, reliability needs, and technician workflows.
- Added safeguards for system folders, junctions, temporary files, caches, and selected AppData paths to reduce migration risk and simplify troubleshooting.

Technical Skills

Languages: Swift, Java, Python, C#, C++, TypeScript, JavaScript, SQL, C

Apple Development: SwiftUI, Swift Concurrency, SwiftData, CloudKit, Vision, Core ML, AVFoundation, Metal

Frameworks & Data: Spring Boot, React, Node.js, .NET 8, PostgreSQL, REST APIs, JPA, Hibernate

Tools & Practices: Xcode, Instruments, OSLog, XCTest, AWS, Docker, Git, Linux, Playwright, Vitest, OOP, Agile/Scrum